

Emerging Science and Technology

Quantum Computing

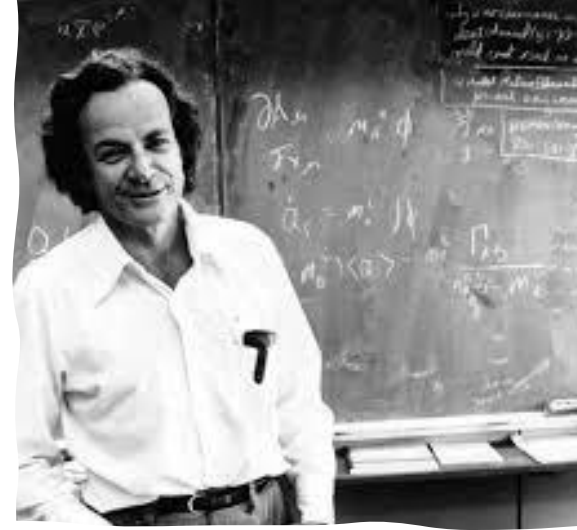
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ES. Learning & Decisions

<https://www.learning-decision.dk>

Start of quantum computing

- Physicists faced an exponential increase in overhead when simulating quantum dynamics
- Yuri Manin and Richard Feynman to independently suggest that hardware based on quantum phenomena might be more efficient for computer simulation
- Current level 50 qubits



Quantum computing – big picture

- 1994: Shor's Algorithm
- 1996: Grover's Algorithm
- Current Applications: Cryptography, Quantum Simulation, Optimization
- Key Challenges: Decoherence, Error Correction, Scalability

Importance

Shor's algorithm is a breakthrough in Cryptography

- Efficiently factorizes large integers.
- Threatens RSA encryption which relies on the difficulty of prime factorization.

Quantum Speedup

- Classical algorithms are exponentially slow for large numbers.
- Shor's Algorithm works in polynomial time → Quantum Advantage.

Grover's algorithm is a quadratic speedup in Search Problems

- Algorithm provides a quadratic speedup for unstructured search problems.

Paves the Way for Quantum Supremacy

- One of the first practical proofs of quantum supremacy.

Current State of Qubits in Quantum Computing

-  Physical Qubits:
 - - IBM Condor Processor: 1,121 qubits
 - - Atom Computing: 1,180 qubits
 - - D-Wave Advantage System: 5,000+ qubits (Quantum Annealing)
-  Logical Qubits:
 - - Quantinuum: 50 logical qubits
 - - Microsoft & Atom Computing: 24 logical qubits

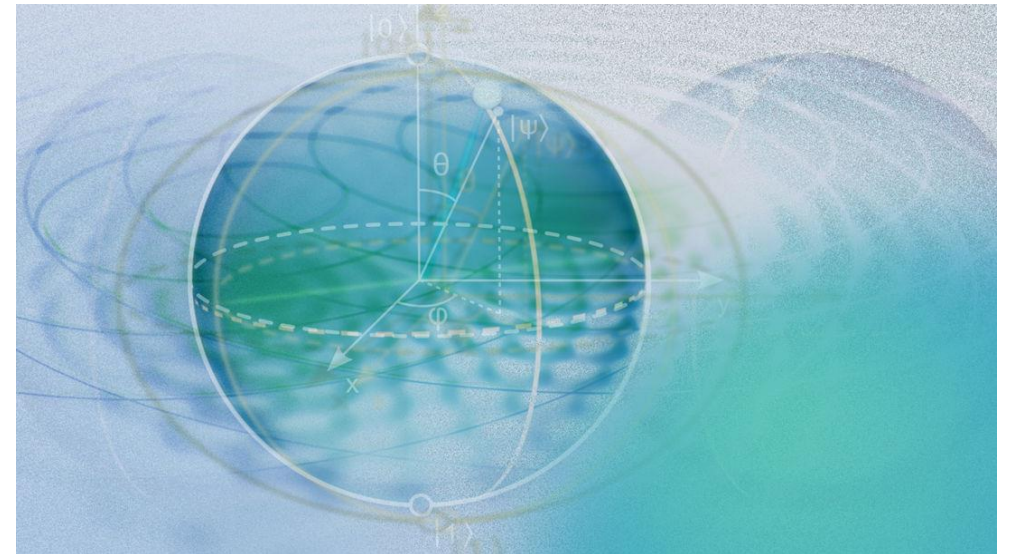
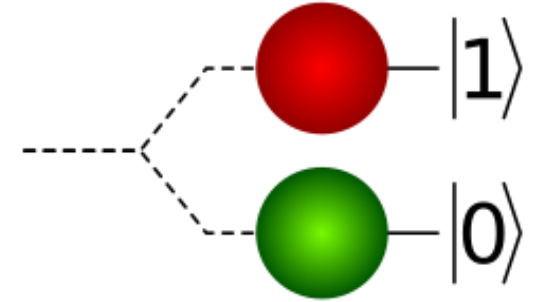
Introduction to Quantum Computing

- Definition of Quantum Computing
- Key Concepts: Qubits, Superposition, Entanglement
- Differences between Classical and Quantum Computing
- Visual Representation of Classical vs Quantum Bits
- Real-world examples (e.g., Quantum teleportation, Quantum Key Distribution)

$$\alpha|0\rangle + \beta|1\rangle$$

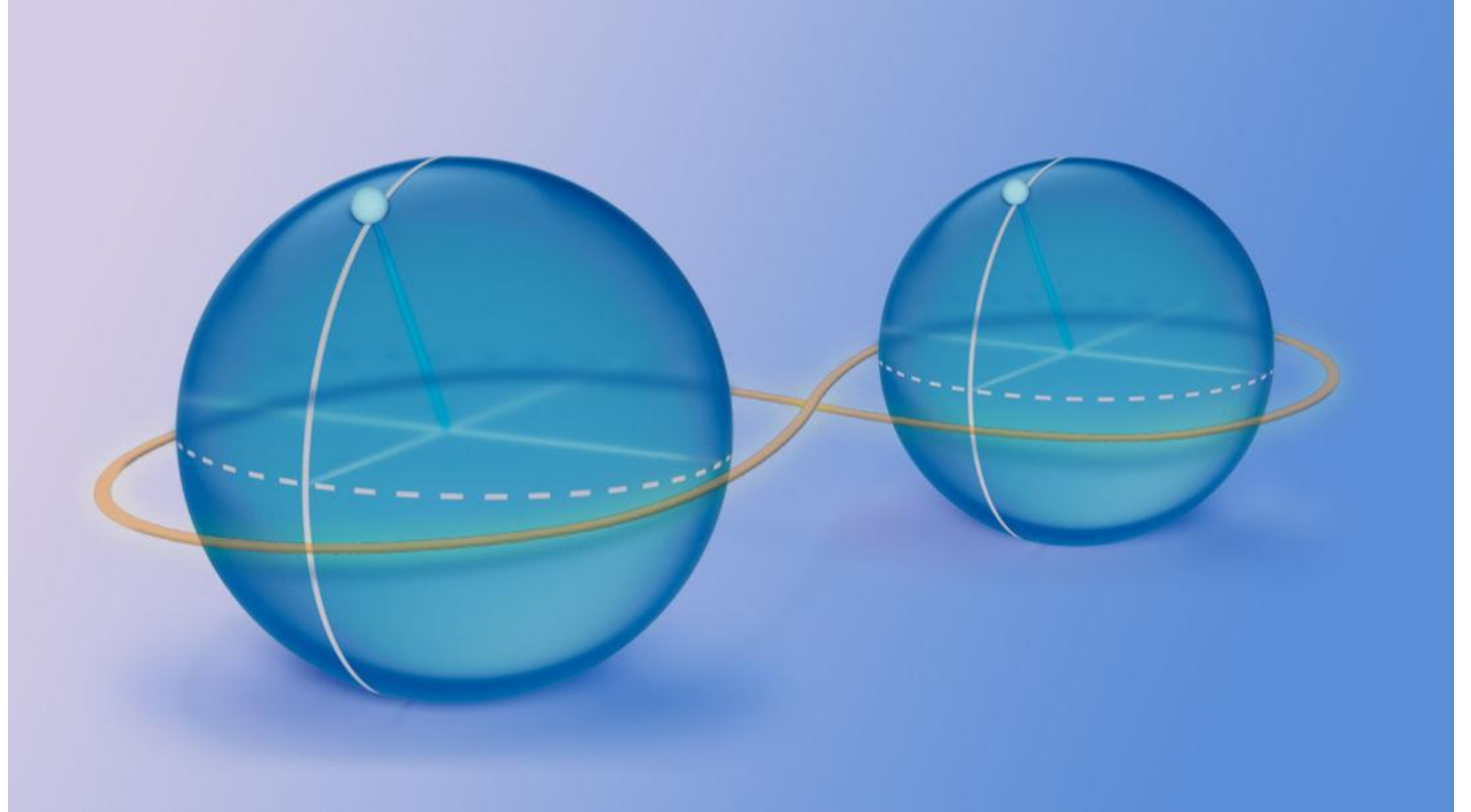
What is a Qubit?

- ◆ A qubit (quantum bit) is the basic unit of quantum information.
- ◆ Unlike classical bits, which are either 0 or 1, a qubit can be both 0 and 1 simultaneously due to superposition.



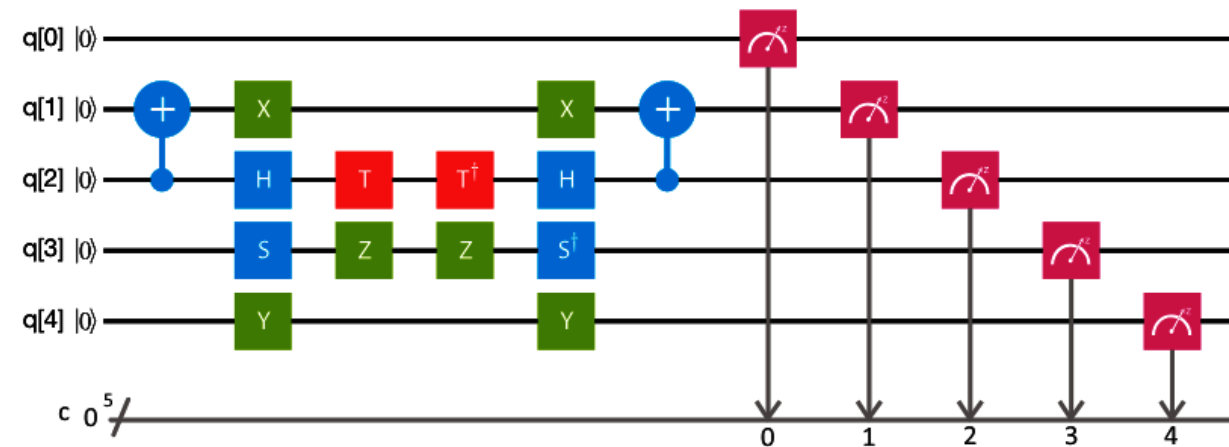
What is a Qubit?

◆ Qubits also exhibit entanglement, allowing the state of one qubit to depend on another, no matter the distance.

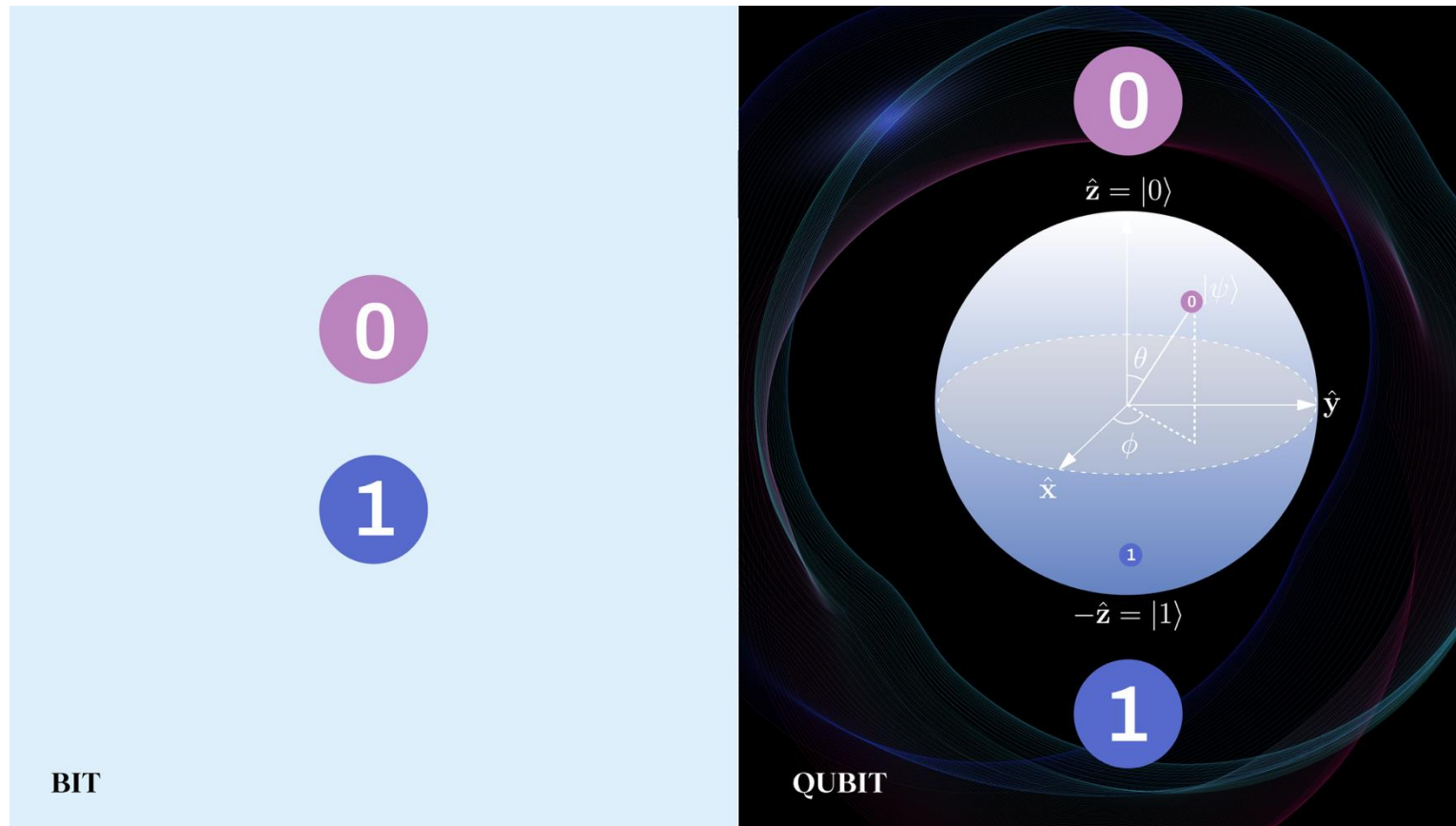


$$\frac{1}{\sqrt{2}} (|0\rangle_A \otimes |1\rangle_B - |1\rangle_A \otimes |0\rangle_B)$$

What is quantum computing



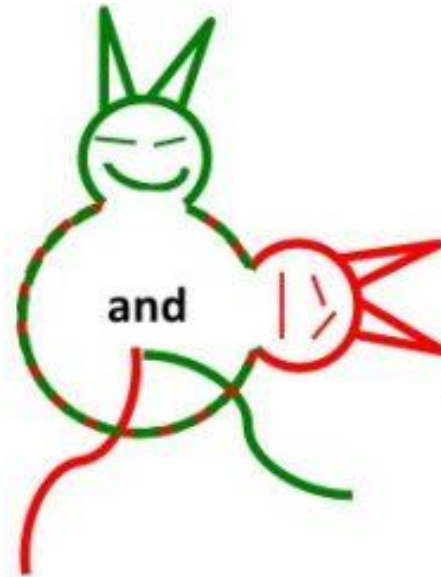
Quantum noise and life-time of a qubit



Quantum measurement

How is the cat?

Quantum physics

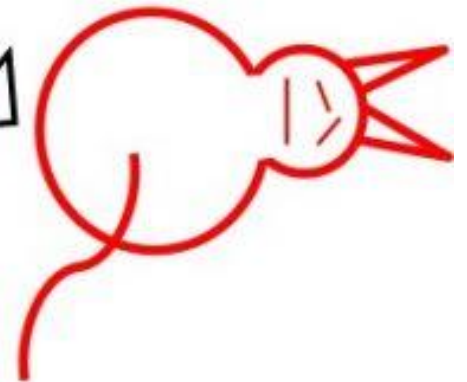


Quantum decoherence

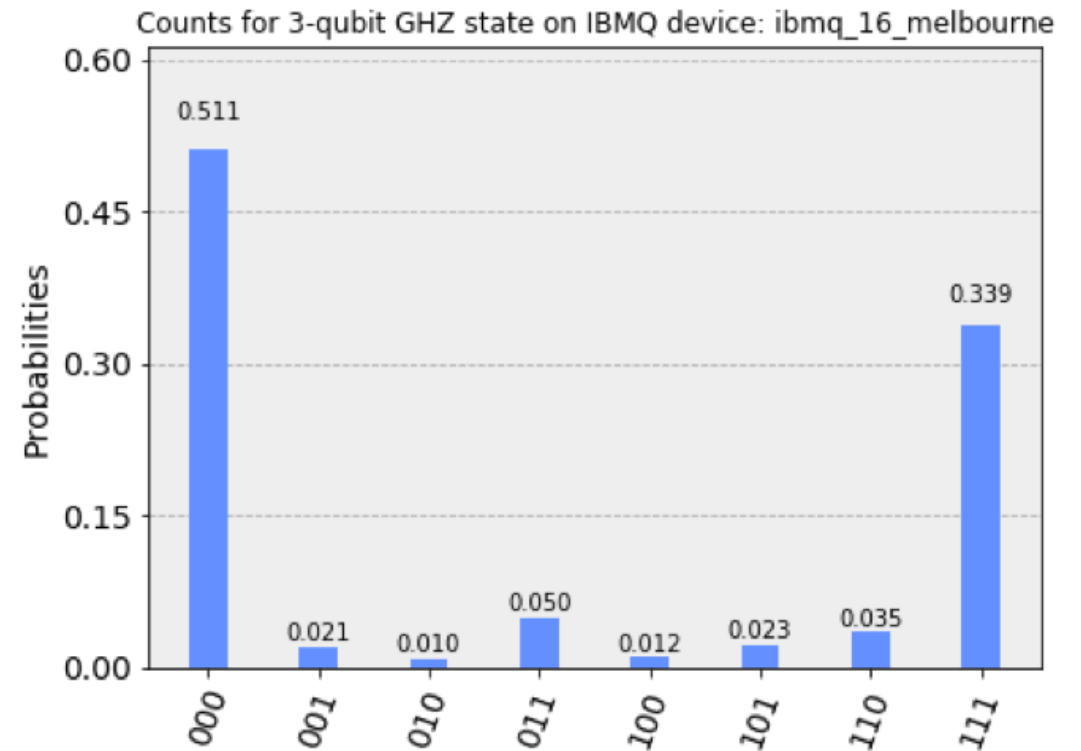
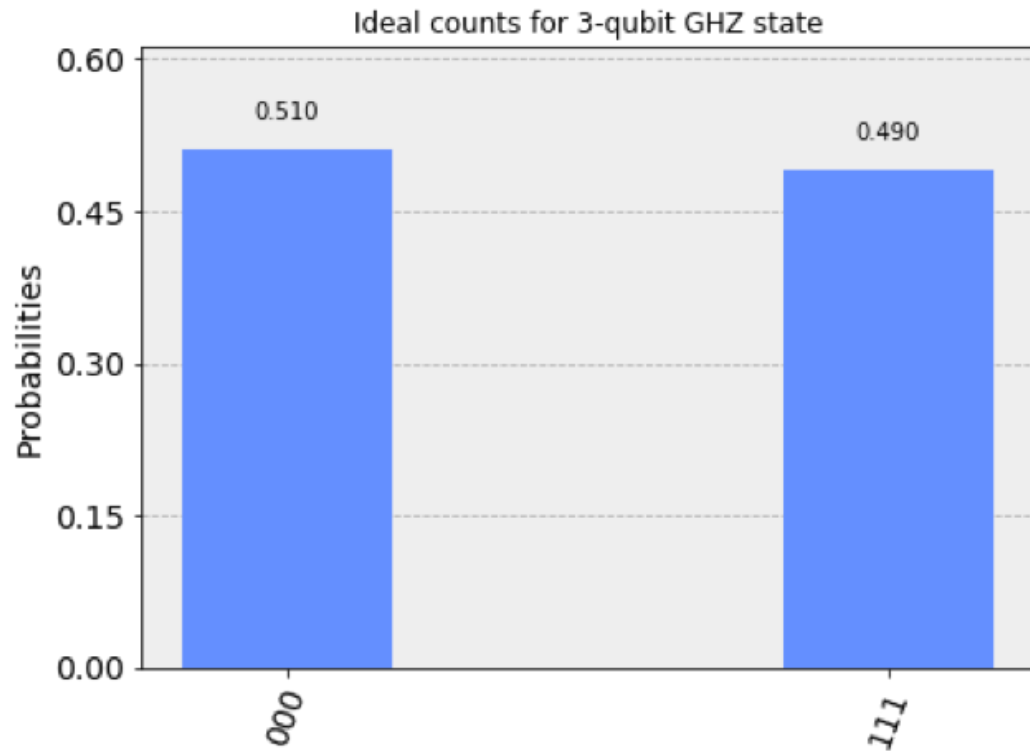
Classical physics



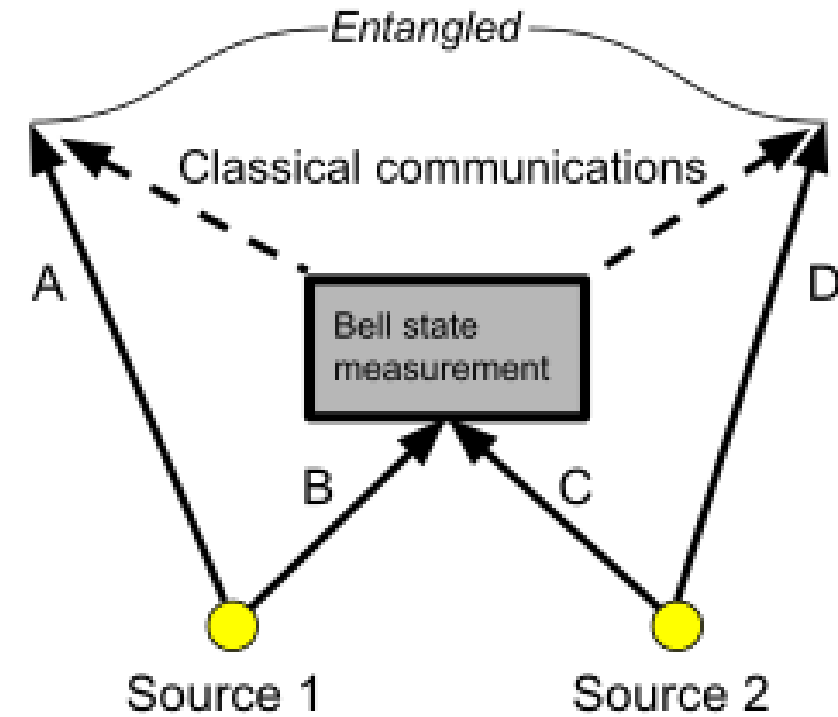
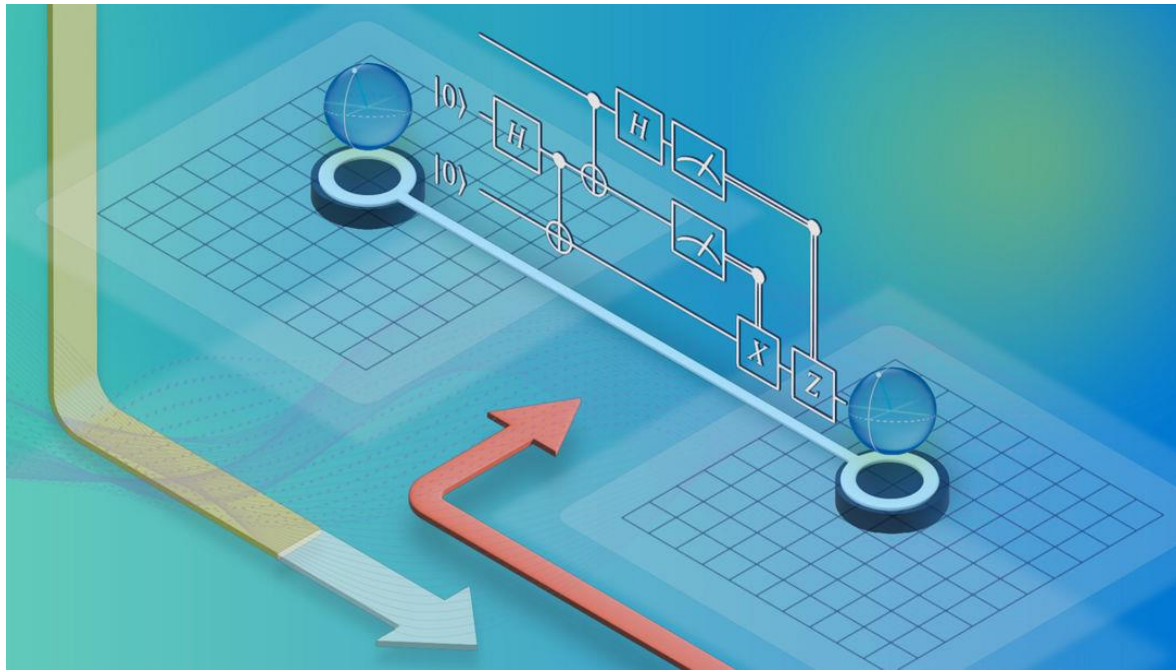
or



Quantum measurements



Teleportation



Key Milestones in Quantum Teleportation

1993 – Theoretical Proposal

Charles H. Bennett and colleagues introduced the concept of quantum teleportation, proposing a method to transmit quantum information using entanglement and classical communication.

1997 – First Experimental Demonstration

Two independent research groups, led by Anton Zeilinger and Francesco De Martini, successfully performed the first quantum teleportation experiments with photons (one meter).

2004 – Teleportation of Atomic Qubits

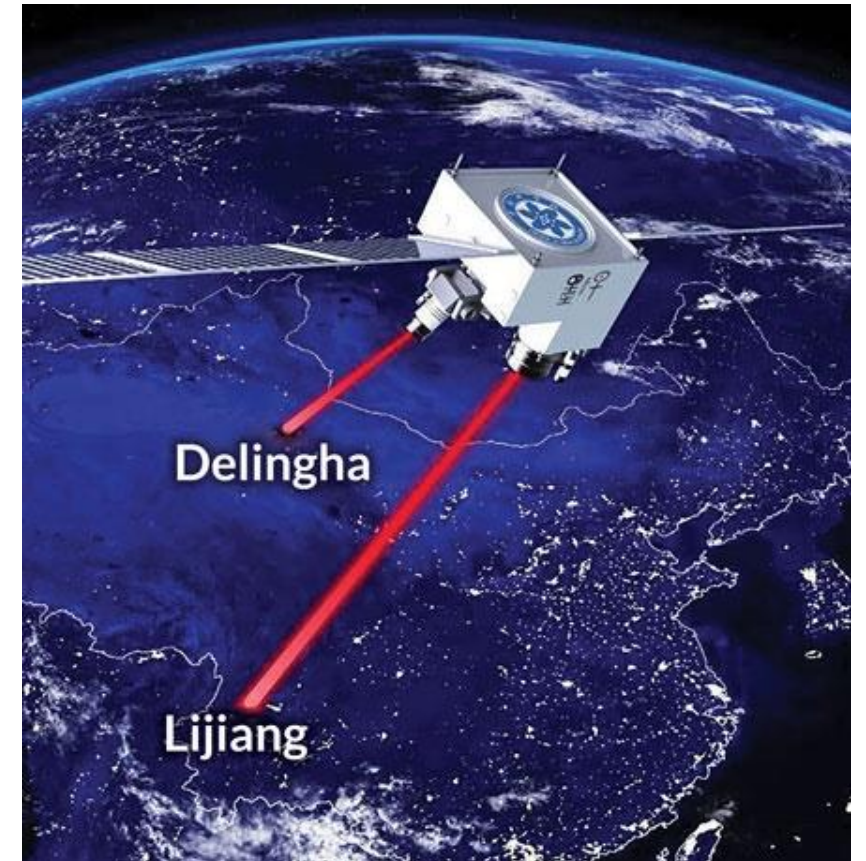
Teams at the University of Innsbruck and the National Institute of Standards and Technology achieved quantum teleportation with trapped ions, marking the first such demonstration with atomic qubits (600 meters across Danube River, Vienna).

2012 – Long-Distance Free-Space Teleportation

Researchers in China accomplished quantum teleportation over a distance of 97 kilometers through free space, setting a new distance record at the time.

2017 – Ground-to-Satellite Teleportation

Chinese scientists successfully teleported quantum states from Earth to the Micius satellite, achieving quantum teleportation over 1,200 kilometers and demonstrating the feasibility of satellite-based quantum communication.



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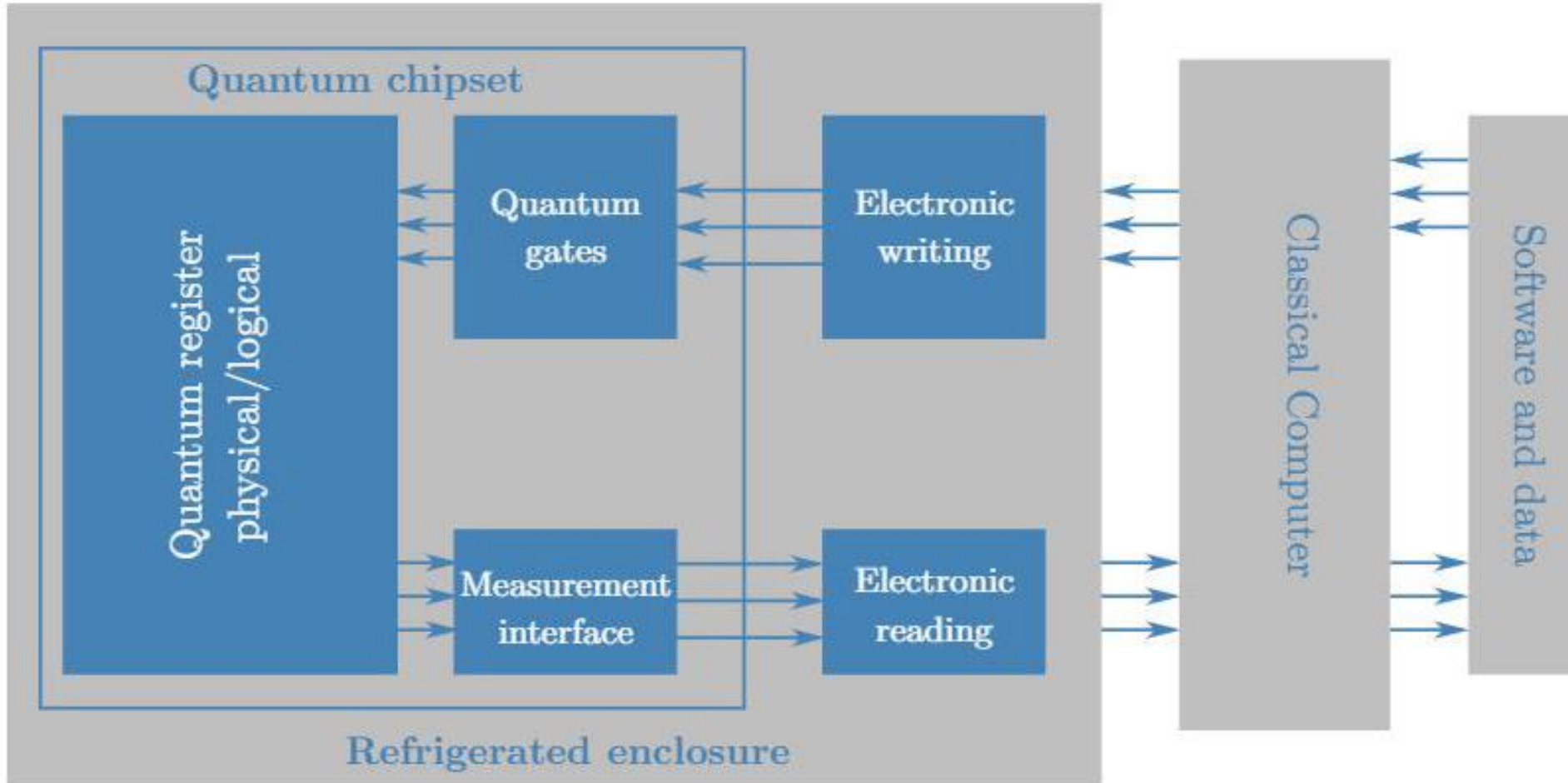
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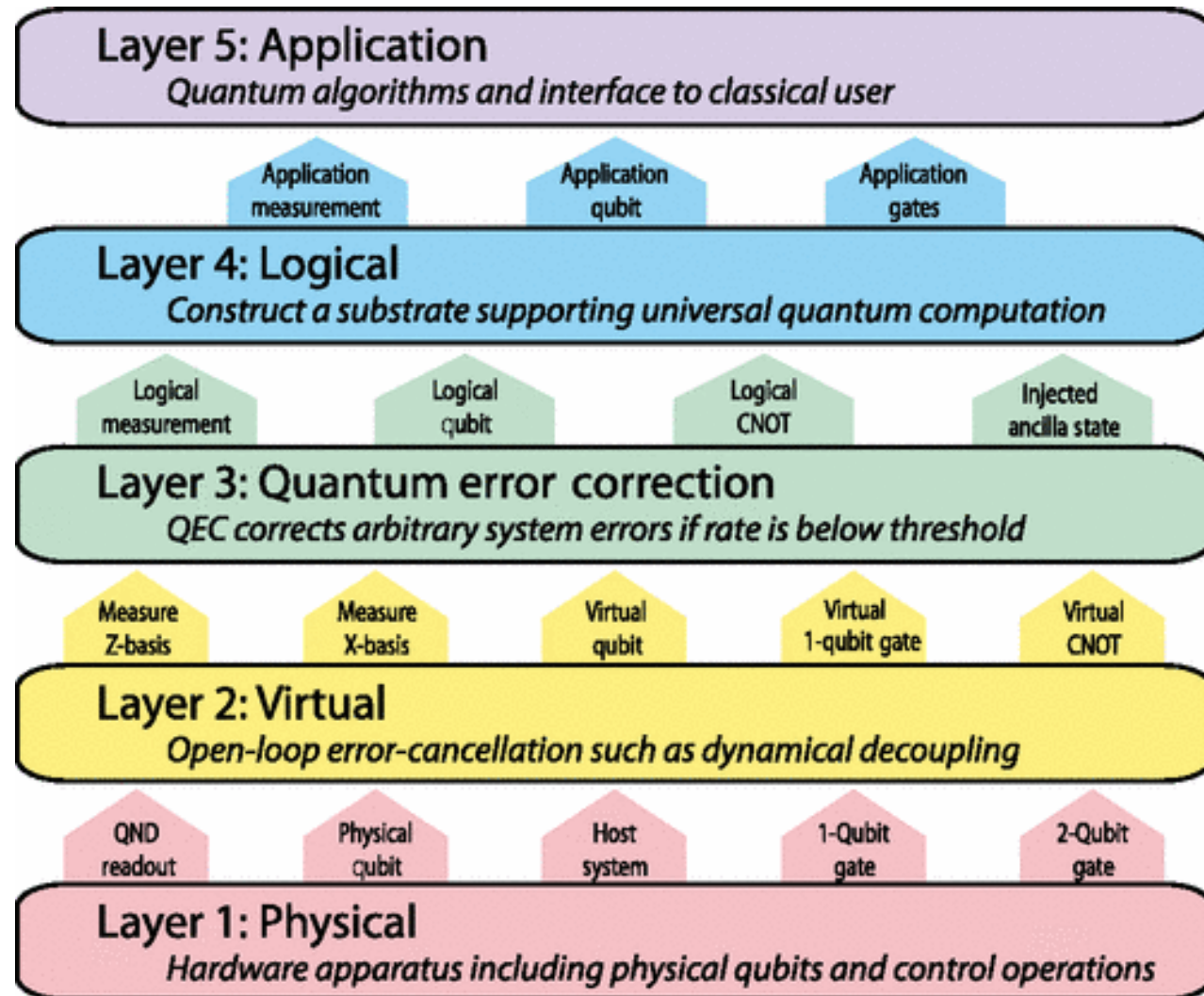
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Quantum Computer architecture



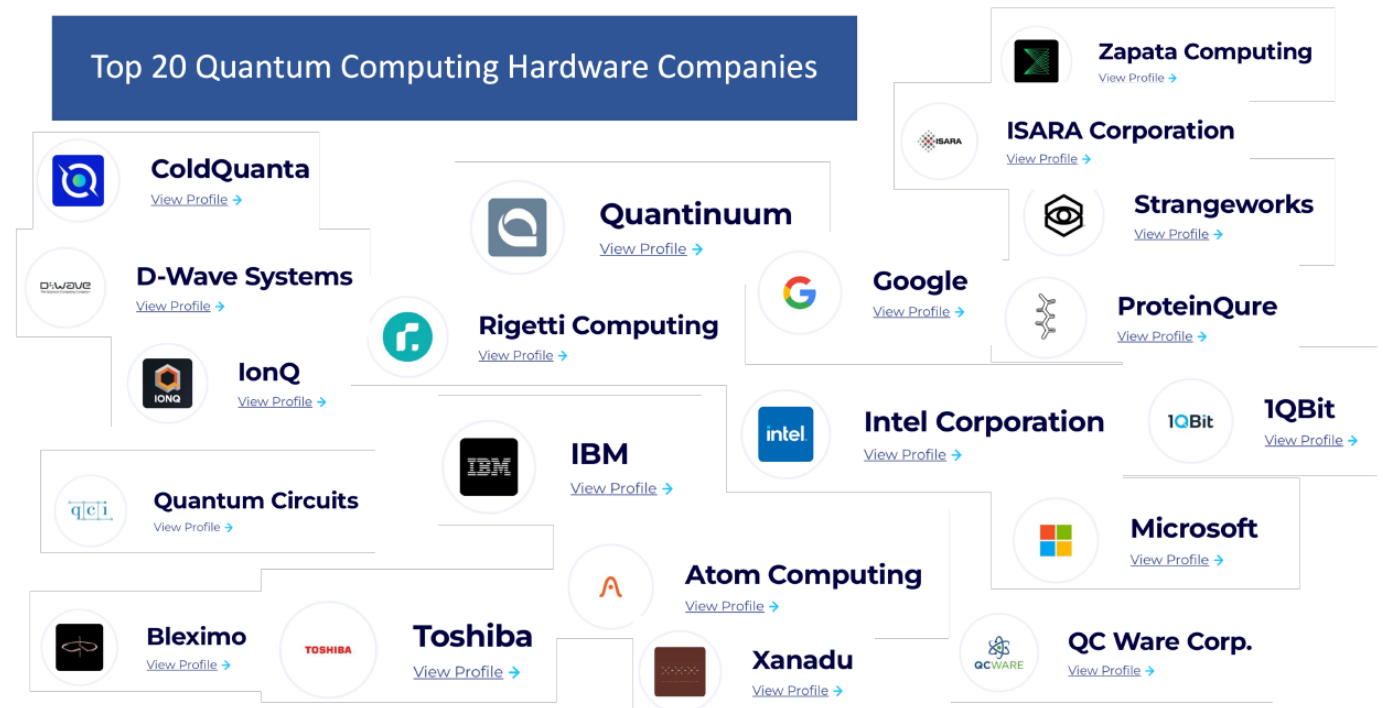
Quantum computer – 5 level stack



Technologies

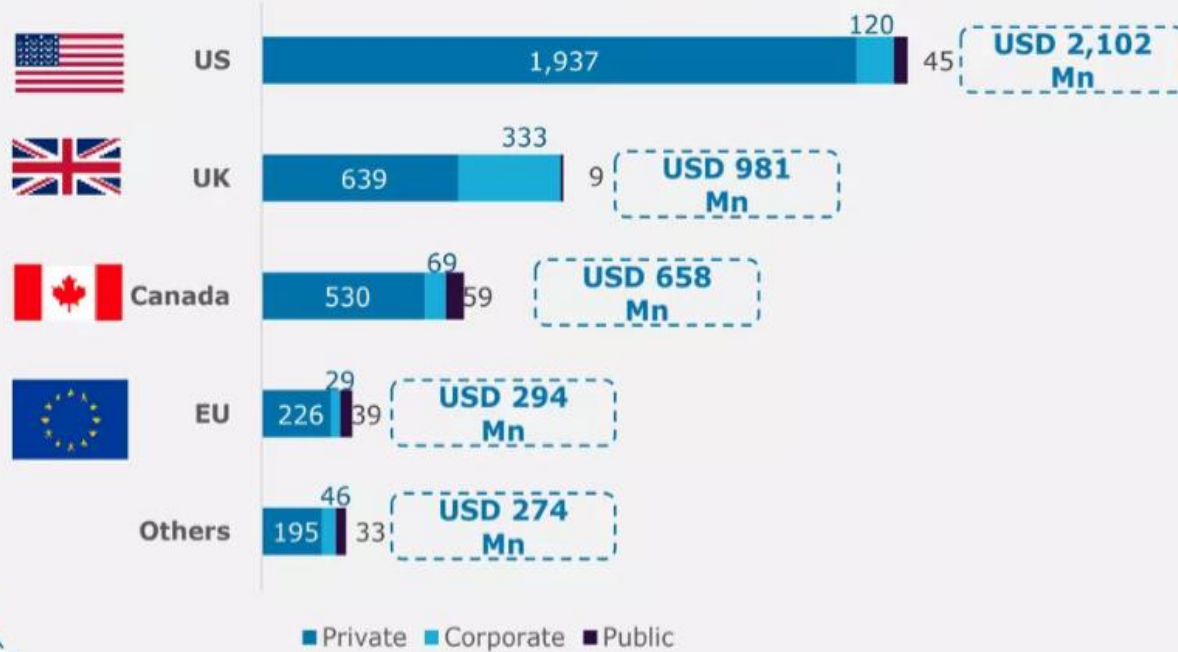
◆ Implemented using different technologies:

- Superconducting circuits
- Trapped ions
- Photonics
- Topological states

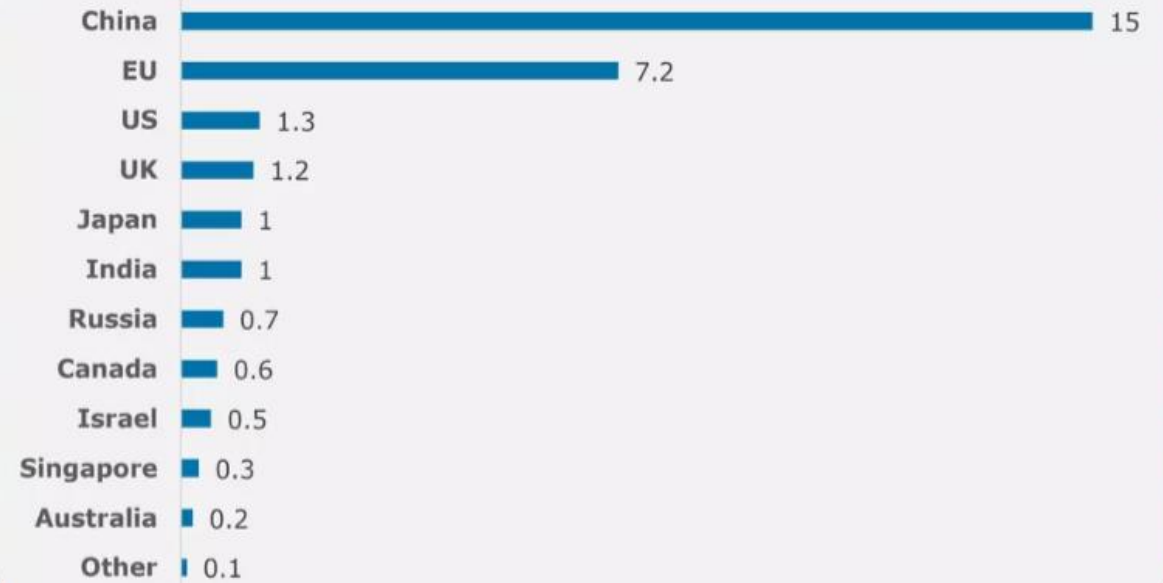


Investments

Global Quantum Investment by Investor Type (2020-2021)



Announced Government Funding (USD Bn)



The European “Quantum Flagship” defines 4 pillars

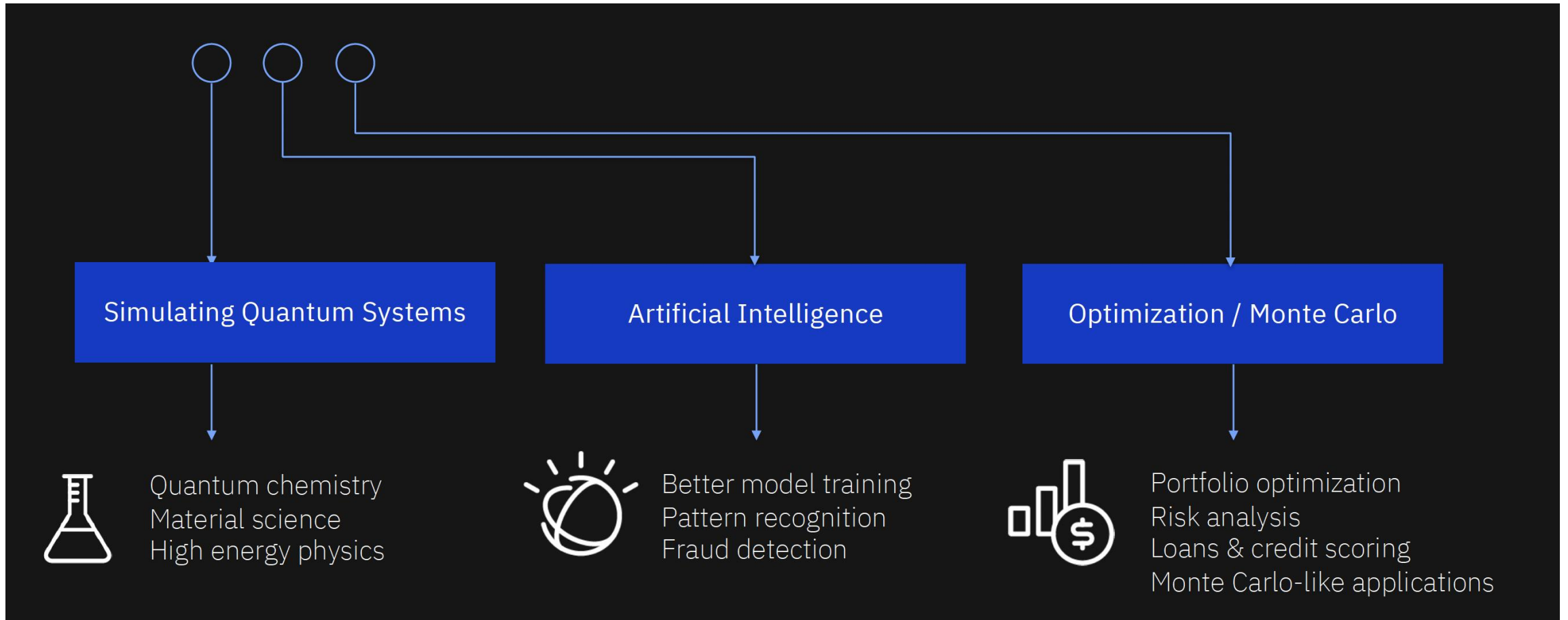
- ◆ *Quantum computing.*
- ◆ *Quantum simulation.*
- ◆ *Quantum communication.*
- ◆ *Quantum metrology and sensing.*

The European Commission launched the major “Quantum Flagship” in 2018 with a minimum lifetime of 10 years, and a budget of at least 1 b€.

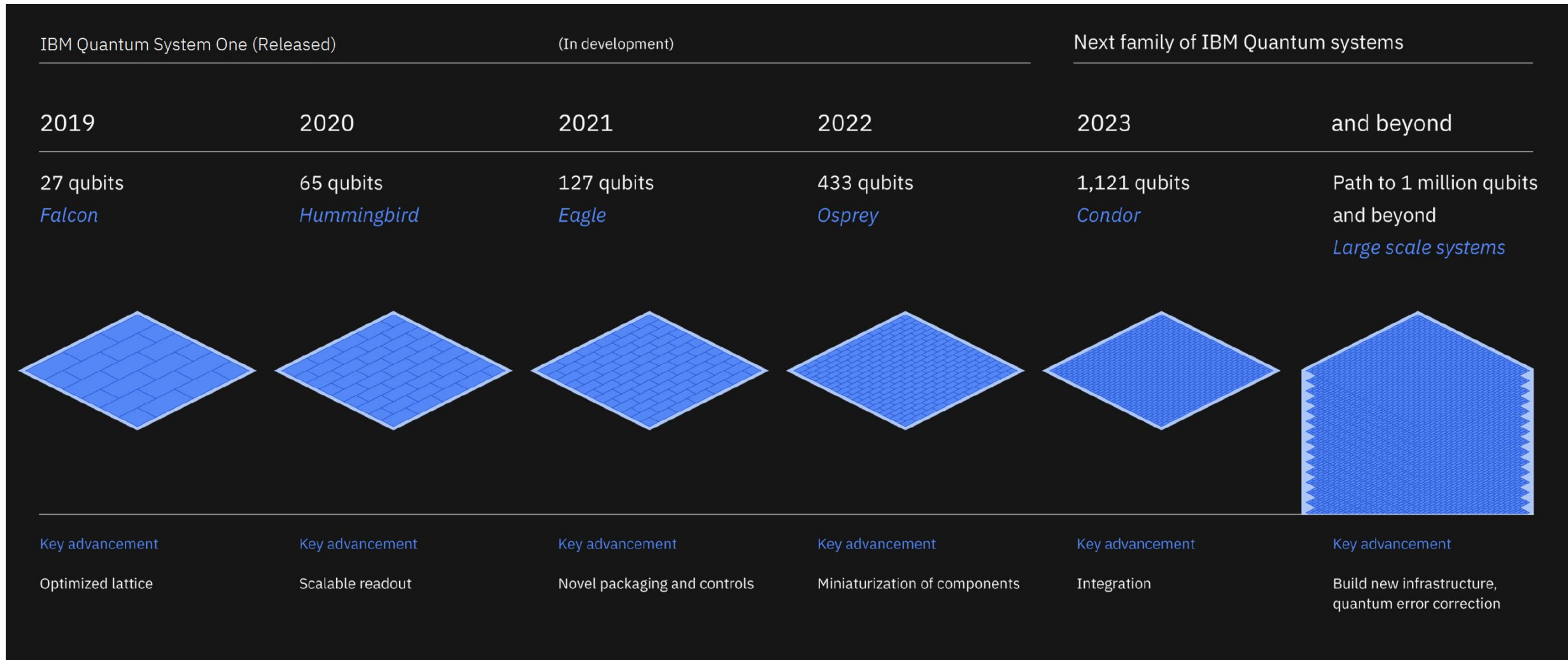
Impacts on Society and Industry

- Cybersecurity and Encryption
- Medicine and Material Science
- Artificial Intelligence and Big Data
- Ethical Considerations and Quantum Governance

Quantum applications



Scaling Quantum Technology by IBM



Quantum Circuits for Applications

Quantum Simulations

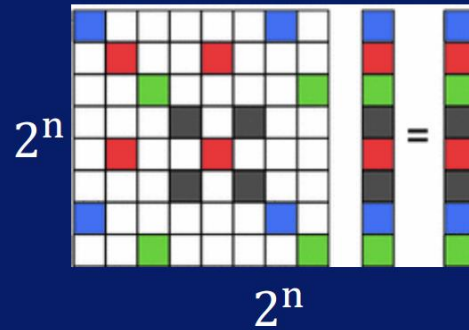


Physics

Chemistry

Materials discovery

Linear Systems ($Ax = b$)



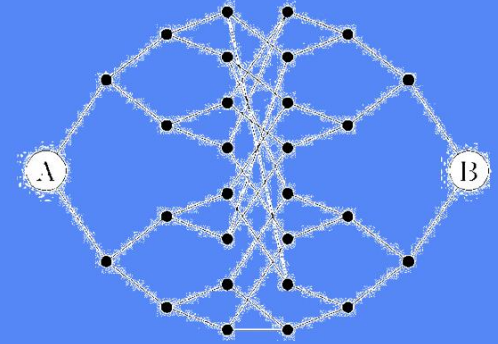
Network analysis

Differential equations

Option pricing, heat transfer

Classification (Machine Learning)

Quantum Walks



Graph properties (network flows, electrical resistance)

Search

Collision finding